

|                                           |                    |                    |
|-------------------------------------------|--------------------|--------------------|
| PRIMARY AI FINDING                        | Model confidence   | 94.9% (calibrated) |
| <b>Meningioma</b>                         | Risk level         | Indeterminate      |
| Borderline - two classes remain plausible | Prediction set     | Glioma, Meningioma |
|                                           | Coverage guarantee | 99%                |

IMPORTANT CAVEATS

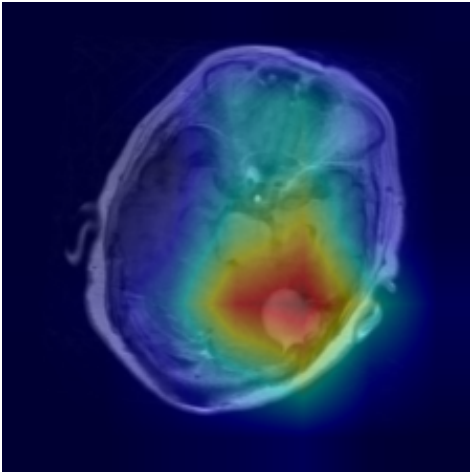
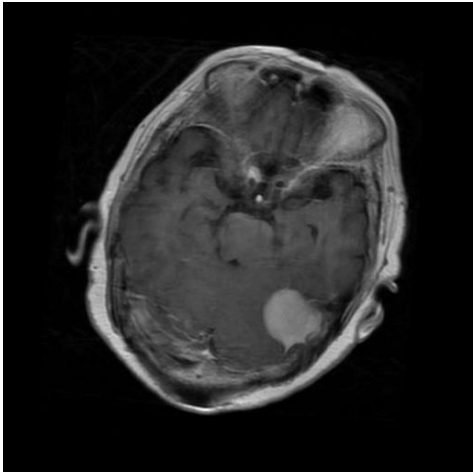
- Risk level was escalated to Indeterminate because the model's uncertainty exceeded the reporting threshold. The class above should not be treated as the answer.

PROBABILITY DISTRIBUTION

| Class      | Probability |             | In set |
|------------|-------------|-------------|--------|
| Meningioma | 94.91%      | <div></div> | yes    |
| Glioma     | 4.99%       | <div></div> | yes    |
| Pituitary  | 0.09%       | <div></div> | -      |
| No Tumor   | 0.01%       | <div></div> | -      |

Probabilities are the stacking meta-learner's posterior. Values are temperature-calibrated. The 'In set' column marks classes retained by the conformal prediction set at the stated coverage level.

EXPLAINABILITY



Original scan (resized)

Grad-CAM attention overlay

The heatmap shows which image regions most influenced the model's output. It is an interpretability aid and must not be read as tumour segmentation, a boundary, or a measurement.

AI SUMMARY

The model classified this scan as 'Meningioma' with 94.9% confidence. A secondary consideration of Glioma (5.0%) was noted but scored lower.

Generated from deterministic templates. No AI-authored narrative has been approved for this finding and uncertainty band.

CONDITION OVERVIEW

Meningioma arises from the meninges, the membrane layers covering the brain and spinal cord. It is a primary central nervous system tumour, meaning it begins in the CNS rather than spreading there from elsewhere. Meningioma is described as the most common primary brain tumour overall, while higher-grade meningiomas are rare.

Presentation varies widely: meningiomas may be found incidentally, or may produce neurological deficits related to tumour location, size, growth rate and histological grade.

Typing and grading follow the WHO CNS classification, 5th edition.

## **LIMITATIONS FOR THIS FINDING**

Meningioma is typically extra-axial and dural-based, which gives it a characteristic appearance ? but the model is not performing localisation or segmentation, and a Grad-CAM heatmap indicates where the network's attention fell, not where a tumour boundary lies.

This project's evaluation showed meningioma with high recall (0.99) but lower precision (0.90) at the single-model baseline, meaning the class over-claims: some gliomas were being absorbed into it. That is the mirror image of glioma's recall shortfall. A meningioma output therefore carries a meaningful probability of being a misassigned glioma, which is exactly why the conformal prediction layer frequently returns the two-class set {Glioma, Meningioma} rather than a single label.

## **POTENTIAL FURTHER EVALUATION**

Investigations a clinician may consider discussing:

- Clinical and neurological examination
- Dedicated contrast-enhanced MRI using a brain-tumour protocol
- Neurosurgical or neurological specialist review where appropriate
- Histological grading where tissue is obtained, since grade influences management and follow-up
- Interval imaging to establish growth rate, which is frequently more informative than a single

study

## **CLINICAL CONSIDERATIONS**

- Many meningiomas are slow-growing, and incidental discovery is common.
- Because presentation depends on location, size, growth rate and grade, two meningiomas of similar appearance can warrant very different management.
- Growth rate over time is a key input and cannot be assessed from one image.
- A single-slice AI classification cannot distinguish grade.

## **FOLLOW-UP CONSIDERATIONS**

- Specialist review to decide whether observation or intervention is appropriate
- Serial imaging where monitoring is the chosen approach, at intervals determined clinically
- Comparison with prior imaging where available
- Reassessment if new or progressive neurological symptoms develop

**WARNING SIGNS - SEEK PROMPT MEDICAL ATTENTION**

Symptoms that generally warrant prompt medical assessment:

- Sudden severe headache, especially if unlike previous headaches
- New seizure activity
- Loss of consciousness
- Sudden weakness, numbness or speech difficulty
- Significant or rapidly progressing visual change
- Confusion or marked change in mental state

General neurological red flags, applicable regardless of AI output.

**TREATMENT INFORMATION (FOR CLINICIAN REVIEW)**

For clinician review. Not a prescription, and no medicines or doses are specified.

Management depends on tumour location, size, symptoms, growth characteristics, histological grade and the patient's overall clinical condition. Broad categories described in the literature include:

- Observation with serial imaging, which is a recognised approach for selected asymptomatic or slow-growing lesions
- Surgical management
- Radiation-based approaches, including stereotactic techniques
- Symptom-directed and supportive care

Selection among these is a specialist decision made after diagnostic confirmation and complete clinical evaluation.

This is general reference information, not a prescription. No medicine, dose or schedule is recommended by this system.

**RECOMMENDED NEXT STEP**

The AI result is inconclusive and should not be acted upon as a finding. Refer this scan for review by a qualified radiologist or specialist physician, who can interpret it alongside the clinical history.

**EVIDENCE AND REFERENCES**

| # | Source                                                                                                              | Publisher / reference                                                                                                                                                                                                                                        |
|---|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Meningioma: International Consortium on Meningiomas consensus review on scientific advances and treatment paradigms | Neuro-Oncology, 2024;26(10):1742-1780 - <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC11449035/">https://pmc.ncbi.nlm.nih.gov/articles/PMC11449035/</a>                                                                                                  |
| 2 | Meningioma (StatPearls)                                                                                             | StatPearls Publishing, via NCBI Bookshelf (NBK560538) - <a href="https://www.ncbi.nlm.nih.gov/books/NBK560538/">https://www.ncbi.nlm.nih.gov/books/NBK560538/</a>                                                                                            |
| 3 | Meningioma: Diagnosis and Treatment                                                                                 | National Cancer Institute (NCI) - Rare Brain and Spine Tumor Program - <a href="https://www.cancer.gov/rare-brain-spine-tumor/tumors/meningioma">https://www.cancer.gov/rare-brain-spine-tumor/tumors/meningioma</a>                                         |
| 4 | Brain and Spinal Cord Tumors                                                                                        | National Institute of Neurological Disorders and Stroke (NINDS), NIH - <a href="https://www.ninds.nih.gov/health-information/disorders/brain-and-spinal-cord-tumors">https://www.ninds.nih.gov/health-information/disorders/brain-and-spinal-cord-tumors</a> |
| 5 | WHO Classification of Tumours of the Central Nervous System, 5th edition                                            | World Health Organization / IARC, 2021 - <a href="https://www.iarc.who.int/">https://www.iarc.who.int/</a>                                                                                                                                                   |

The clinical content in this report is pending verification against the primary sources by a qualified reviewer.

**AI LIMITATIONS AND INTENDED USE**

*This AI-generated report is for clinical decision support only. It must not replace professional medical diagnosis. All findings should be validated by a qualified radiologist or specialist physician.*